

Frogs and toads

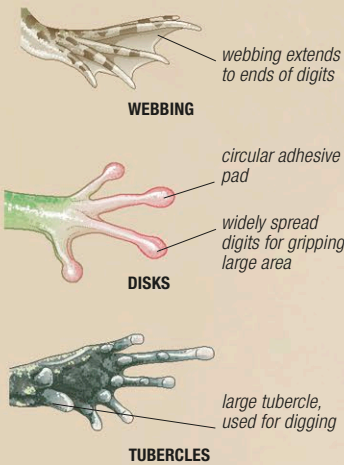
PHYLUM	Chordata
CLASS	Amphibia
ORDER	Anura
FAMILIES	56
SPECIES	6,700

Frogs and toads form much the largest and best known of the 3 groups of amphibians. As adults, they have no tail—unlike other amphibians, their tail is absorbed during metamorphosis from the larval to the adult stage. There is no clear distinction between frogs and toads: the term

“toad” is often used to refer only to members of the genus *Bufo*, but it is also used more widely to describe any slow-moving, terrestrial species with a squat body and rough, warty skin. The larvae of frogs and toads, known as tadpoles, mainly feed on algae or plant material and characteristically have a spherical body that contains the long, coiled gut required by such a diet. In many adults, the back limbs are modified for jumping; they are much longer and more muscular than the front limbs. Most frogs and toads live in damp habitats, near the pools and streams in which they breed, but there are several species that live in arid areas. The greatest diversity of species is found in the tropics, particularly in rain forests.

Anatomy

Frogs and toads have a short, rigid body, a large, wide head, and, in most species, front limbs that are much smaller than the back limbs. The back limbs contain an extra section, just above the foot, that is not present in other amphibians. Most species have large, protruding eyes and a conspicuous eardrum (tympanum) on each side of the head, reflecting the importance of the senses of vision and hearing. The mouth is large, and many species have an adhesive tongue that they can flick out at high speed to catch their prey. The female is usually larger than the male, but males of many species have thicker and more muscular front limbs. These enable the male to grip the female securely during mating.



FEET
Frogs and toads have 4 digits on the front limbs and 5 on the back. In many species, especially those that spend much of their life in water, the digits on the limbs have webs of skin between them. Many frogs that live in trees have disk-shaped adhesive pads on the ends of their digits, which enable them to gain a secure foothold on smooth, vertical surfaces. Burrowing species have horny protuberances, called tubercles, on their back feet, which they use to excavate soil.

SKIN
The skin of frogs and toads is highly variable. At one extreme, the skin of some toads is so tough that it is used as a substitute for leather; at the other, there are frogs with skin so thin that their internal organs can be seen through it. Toads typically have rough, dry skin that is covered with warts and, in some species, small spines. Frogs generally have smooth, moist skin.



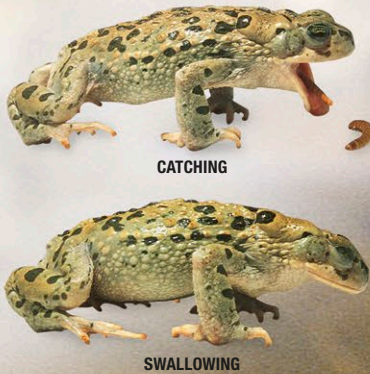
FROG SKIN



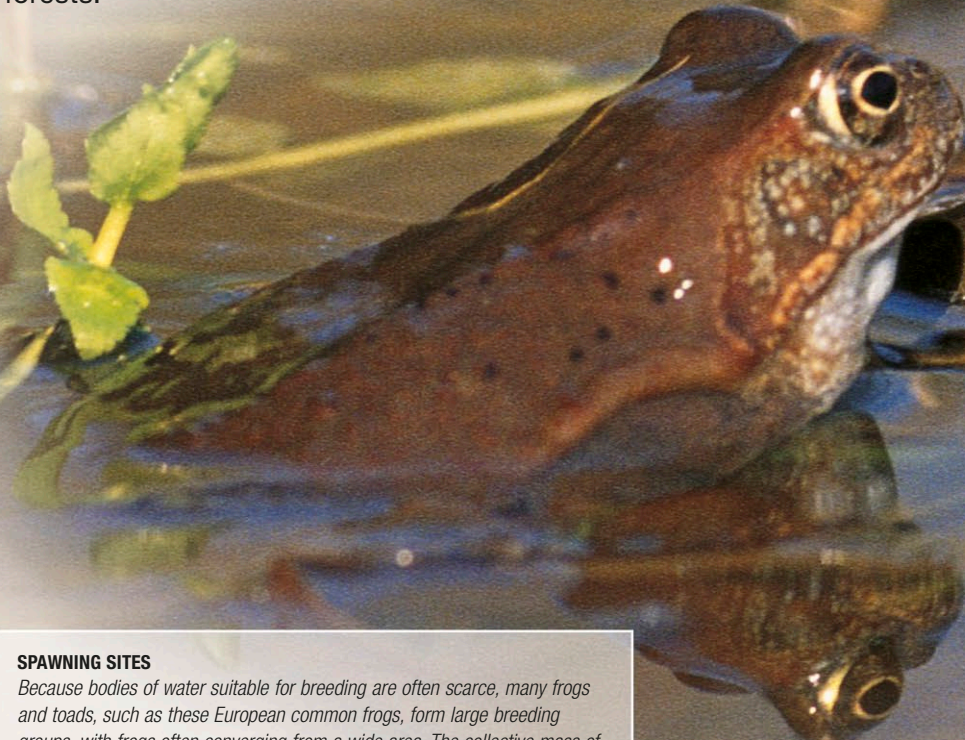
TOAD SKIN

Feeding

All frogs and toads are carnivores that feed on live prey rather than carrion. Unable to chew or break up food in their mouth, they have to swallow their prey whole. However, many species can eat large animals (including mice, birds, and snakes), and a single meal can meet their energy needs for a long time so that they do not need to feed often. Few species are active hunters, the most common feeding method being to wait until prey comes within range, either of their long, sticky tongue or of a forward, open-mouthed lunge.



SWALLOWING FOOD
Frogs and toads typically shut their eyes when swallowing their food. As their eyelids close, the eyeballs are rolled and pushed downward to increase pressure in the mouth.



SPAWNING SITES
Because bodies of water suitable for breeding are often scarce, many frogs and toads, such as these European common frogs, form large breeding groups, with frogs often converging from a wide area. The collective mass of eggs is also able to retain more heat, enabling the tadpoles to hatch sooner. Ponds and streams that dry up for part of the year are often used, since they do not support other animals that eat eggs and tadpoles.

FINDING A MATE
Many male frogs and toads call to attract females to the spawning site. The calls made by males enable a female to find a mate. More importantly, each species has its own characteristic call, which helps a female to find a mate of the same species in breeding pools used by several species at once. In many species (such as the aquatic frog, right), the effectiveness of the call is enhanced by one or more vocal sacs. The male fills himself with air, which is moved back and forth between the lungs and vocal sacs; sound is produced as it passes over the vocal cords in the larynx.

